Math 8/9 - Statistics and Probability



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

Copyright © 2024 by **Douglas Tam**

All rights reserved.

No portion of this book may be reproduced in any form without written permission from the author, or **St. George's School**.

Last Updated: May 1, 2025

Contents

1 Central Tendency **4**
- The Mean . . . 4
- The Median . . . 4
- The Mode . . . 5
- Finding a Value Given the Mean . . . 5
- Practice Exercises . . . 6

2 Collecting Data and Biases **9**
- Influencing Factors . . . 9
- Census vs Samples . . . 12
- Sampling Methods . . . 13
- Practice Exercises . . . 15

3 Representing Data **19**
- Different Types of Data . . . 20
  - Categorical Data . . . 20
  - Numerical Data . . . 20
- Types of Graphs . . . 21
- Constructing a Circle Graph . . . 21
- Practice Exercises . . . 22

4 Misrepresenting Data **25**
- Scaling . . . 26
- Practice Exercises . . . 27

5 Probability **29**
- Probability Notation . . . 31
- Complement Events . . . 31
- Sample Space . . . 32
- Independent and Dependent Events . . . 33
- Practice Exercises . . . 35

6 Reasoning and Analyzing Problems **38**
- Practice Exercises . . . 39

Answer Key **40**
- Section 1 - Exercises . . . 40
- Section 2 - Exercises . . . 40
- Section 3 - Exercises . . . 42
- Section 4 - Exercises . . . 43
- Section 5 - Exercises . . . 44
- Section 6 - Exercises . . . 45

1 Central Tendency

The Mean

When discussing about the "center" of data, the most common measure is the **average** or **mean**.

The **mean**, $\overline{x}$, of a data set is the statistical name for its arithmetic average. It is found by dividing the sum of the data values by the number of data values.

$$\overline{x} = \frac{sum\,of\,data\,values}{number\,of\,data\,values}$$

Example 1.1 Find the mean of the following data set:

7 10 12 10 8 13 10 6 5 11 6 10

The Median

The **median** is the *middle value* of an **ordered data set**. If there are n values, then the median is the $\left(\frac{n+1}{2}\right)^{th}$ data value.

If n is even, then there is no individual middle term. Then we take the average of the two terms nearest the middle.

Example 1.2 Find the median for each set.

a) $3, 6, 5, 7, 7, 4, 6, 5, 6, 7, 6, 8, 10, 7, 8$

b) $3, 6, 5, 7, 7, 4, 6, 5, 6, 7, 6, 8, 10, 7, 8, 9$

The Mode

The **mode** is the number that shows up the most frequent. There can be no modes if no data number occurs more than another, or there can be multiple modes if multiples numbers appear equally.

Example 1.3 Find the mode for the following data set.

a) $3, 6, 5, 7, 7, 4, 6, 5, 6, 7, 6, 8, 10, 7, 8, 9$

Finding a Value Given the Mean

If you know the mean value, you know the total sum. With the total sum, we can work backwards to solve for certain data.

Example 1.4 On your first 7 tests, you averaged 85%. You have one more test coming up and you want to increase your final average to 86%. What mark do you need on your next test to achieve this?

Example 1.5 The mean of five numbers is 12. Four of the numbers are 10, 15, 8, and 14. What is the unknown number?

Practice Exercises

1. Determine the mean, median, and mode for the following lists of numbers. Round to the nearest tenths where necessary.

   a) $2, 3, 3, 3, 4, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6, 7, 7, 8, 8, 8, 9, 9$

   b) $10, 12, 12, 15, 15, 16, 16, 17, 18, 18, 18, 18, 19, 20, 21$

   c) $22.4, 24.6, 21.8, 26.4, 24.9, 25.0, 23.5, 26.1, 25.3, 29.5, 23.5$

   d) $127, 123, 115, 105, 145, 133, 142, 115, 135, 148, 129, 127, 103, 130, 146, 140, 125, 124, 119, 128, 141$

2. The annual salaries of ten office workers are:

   $$\$33000, \$56000, \$52000, \$48000, \$38000$$

   $$\$45000, \$33000, \$48000, \$33000, \$42000$$

   a) Find the mean, median, and modal salaries of this group.

   b) Explain why the mode is an unsatisfactory measure of the middle in this case.

3. It started raining on Sunday and 11 mm of rain fell. On the next 5 days: 24 mm, 28 mm, 22 mm, 16 mm, and 13 mm of rain fell.

   a) Find the mean rainfall from Sunday to Friday.

   b) How much rain would need to fall on Saturday for the mean to remain the same?

   c) If 10 mm of rain fell on Saturday, calculate the mean rainfall for the week.

4. A small business has 10 people in total on the payroll.

   - Eight workers making $10000 per year.
   - One foremean making $40000 per year.
   - One owner making $880,000 per year.

   a) Determine the mean, median, and mode for the 10 people.

   b) If you were the owner, what type of average would you prefer to use in wage negotiations? Why?

   c) If you were a worker, what type of average would you prefer to use in wage negotiations? Why?

5. Answer the following.

   a) What form of central tendency is always part of the data?

   b) What form of central tendency requires using all values of the data?

   c) What form of central tendency is the centre or middle value of the data?

   d) What form of central tendency is least likely to be an actual score of the data?

6. If there are 8 numbers with a mean of 10, and 12 other numbers with a mean of 16, what is the mean of all 20 numbers?

7. The mean of 50 numbers is 18. If the mean of the first 30 numbers is 16, what is the mean of the last 20 numbers?

8. The mean of 50 numbers is 38. If the numbers 45 and 55 are removed from the set of 50 numbers, what is the mean of the remaining numbers?

2 Collecting Data and Biases

Data collection can be influenced by many different factors. Things like bias, timing, wording, privacy, cultural sensitivity are among some influencing factors. Often, we collect data by ways of surveys, and we want to create survey questions that do not influence the data collected.

Example 2.1 A customer walks into an electronics store, the salesperson immediately approaches the customer and asks if they're here to look for a new appliance or some mobile electronics, or both.

a) Why do you think the salesperson is asking this question?

b) Is this data collection important information for the owner of the store?

c) Are there different ways we can collect this information?

Influencing Factors

Bias	When the whole survey favours one side of data collection.
Word Choices	When specific words in the question may sway answers.
Timing	*When* you collect data can change the results (like asking about snow tires in summer).
Privacy	People might not answer honestly (or at all) if questions are too personal. No-name surveys help.
Cultural Awareness	Being respectful of different backgrounds and not asking questions that might offend or exclude people.
Ethics	Collecting info fairly - being honest, respecting people, and not tricking them or causing harm.
Cost	How expensive it is to gather the information (printing surveys, paying staff, etc.).
Time	How long it takes to collect all the needed information.

Example 2.2 For each situation, identify any **influencing factors.**

a) A fast-food chain emails a customer satisfaction survey immediately after a purchase, offering a free dessert for completing it. The question read "How satisfied were you with your excellent meal?"

b) A university sends out a diversity and inclusion survey to students but only in English, despite having many international students.

c) A tech company tests a new AI chatbot by releasing it to only young adults in urban areas for feedback.

d) A hospital asks patients to fill out a mental health survey while they wait in the emergency room.

e) A political campaign uses social media analytics to tailor ads based on users' browsing history without explicit consent.

Example 2.3 Explain why a problem might occur and the effect it would have on the results. Suggest how each problem can be avoided.

a) A company sends a "workplace satisfaction" survey to employees right after announcing annual bonuses.

b) A hospital asks Muslim patients to rate their satisfaction with care during Ramadan, without offering after-sunset options.

c) A retail app asks users to rate products, then publicly shares their responses with their full name unless they manually opt out.

d) A startup tests a new product only with wealthy urban customers to save on rural shipping costs.

Example 2.4 Rewrite the following question free of influencing factors.

"Should we ban smartphones entirely to protect learning, or let students get distracted all day?"

Census vs Samples

Population: The entire group you want to study (e.g., all Grade 9 students in your school). Census: Collecting data from every single member of the population.

When to Use a Census

- Pros: Super accurate!
- Cons: Expensive and time-consuming for large groups (e.g., surveying every Canadian for Stats Canada).

Example 2.5 Why does Stats Canada do a census every 5 years? What household info might they ask for? (Hint: Think taxes, healthcare, or schools!)

When to Use a Sample

- Pros: Faster, cheaper, and often just as useful!
- Cons: May not get accurate representation.

Example 2.6 To survey Grade 9 students about attendance policies, the student government asks 5 out of 13 Grade 9 classes. Why a sample? Would polling every student take too long? Will this provide an accurate representation?

Sampling Methods

There are multiple sampling options that we can take. Here are some common ones.

Simple Random Sampling

- How: Like drawing names from a hat!
- Example: Pick 5 random students from your math class by assigning numbers and using a random number generator.

Systematic Sampling

- How: Pick every n^{th} person (e.g., every 10th student in the attendance list).
- Example: A factory tests every 50th granola bar to check quality.

Cluster Sampling

- How: Randomly pick whole groups (e.g., one grade level), then survey everyone in that group.
- Example: Survey all Grade 10 students (but only Grade 10).

Self-Selected Sampling

- How: Only volunteers respond (warning: biased!).
- Example: A YouTube poll asking,"Should homework be banned?" (Only people with strong opinions click!).

Convenience Sampling

- How: Ask whoever's easiest (warning: often biased!).
- Example: Surveying only students in the cafeteria about lunch preferences.

Stratified Random Sampling

- How: Randomly pick some people from each subgroup (e.g., 5 students per grade).
- Example: Poll 3 students from each Grade 9, 10, 11, and 12 for a school-wide opinion.

Example 2.7 ***Café Hours Survey***
The student government wants to know if students would support extending café hours. Evaluate these sampling methods:

a) **Random Selection**
Every student's name is entered in a lottery system, and 100 are randomly chosen.

b) **Systematic Sampling**
Every 5th student entering school is surveyed between 8:00-8:30 AM.

c) **Friend Network**
Student government members only survey their own friends.

d) **Café Patrons**
Only students currently buying café items are surveyed.

Example 2.8 ***Choosing Appropriate Samples***
A company packages boxes of granola bars. The quality-control manager inspects the first 5 boxes each morning to ensure that each has the same number and types of granola bars.

a) Is this a good way of ensuring quality control?

b) Suggest 2 other methods of sampling that would be appropriate.

Practice Exercises

1. In each case, describe the effect of influencing factors on data collection:

   a) A sales representative asks: *Which do you prefer?*

   A. Cola B. Diet Cola

   b) An opposition party member asks: *Is the current prime minister not the best in Canadian history?*

   YES NO

   c) Rob asks students: *What is your parents' total income?*

2. Identify one problem with each data collection method:

   a) After the first week of school, your principal asks only you and your friends about enjoying school.

   b) An online magazine forces readers to choose: *"If you find $20, you turn it in."*

   Agree Disagree

   c) Brenda asks classmates: "Should girls be banned from covering their heads in school?"

   d) Carlos plans to ask every student their favourite movie type to determine school-wide preferences.

3. Explain how each question reveals the questioner's bias:

   a) "Should DNA tests convict violent criminals?"

   b) "Should gas-guzzling SUVs be banned?"

   c) "Should students use spell check since it automatically improves spelling?"

4. Rewrite the previous questions to eliminate bias and explain improvements:

   a)

   b)

   c)

5. List 3 topics people might prefer to answer anonymously:

6. Identify whether each uses a **sample** or **census**:

   a) To determine Grade 9 students' favorite TV show, **all** Grade 9 students are surveyed.

   b) To assess customer satisfaction at a coffee chain, **some** customers in every shop were surveyed.

7. Name the target population for each opinion survey:

   a) Bus fares

   b) Goods and Services Tax (GST)

   c) Daycare costs

   d) Emergency room wait times

8. Identify one sampling problem in each scenario:

   a) Surveying only mall shoppers about whether "most people enjoy shopping"

   b) School cook surveys only teachers about cafeteria menu items

   c) Fitness magazine collects teen exercise data only from its readers

9. Evaluate each sampling method's appropriateness:

   a) TV show asks viewers about town speed limits

   b) Restaurant polls every 8th customer on one day

   c) School randomly selects 50 student IDs for policy feedback

   d) 10 students per grade surveyed about cycling habits

10. Propose appropriate sampling methods:

    a) Prime Minister needs **national budget opinions**

    b) School newspaper predicts **election results**

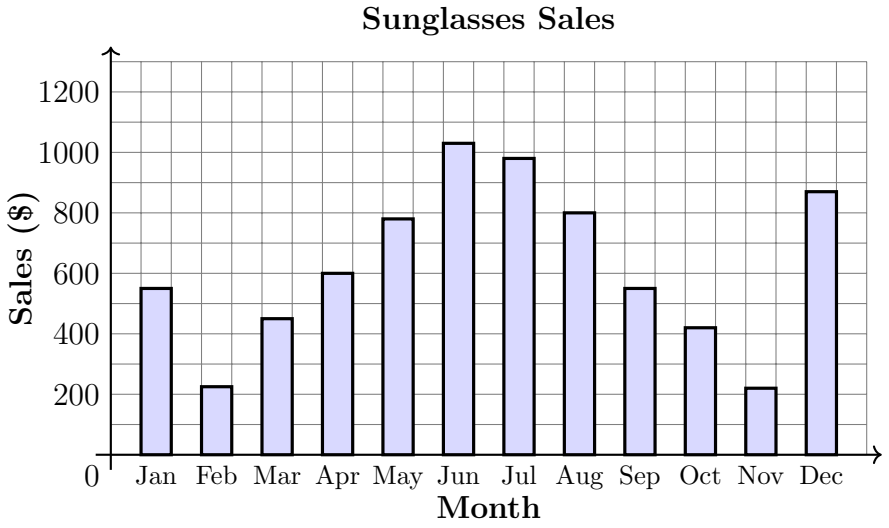
11. Critical Thinking:

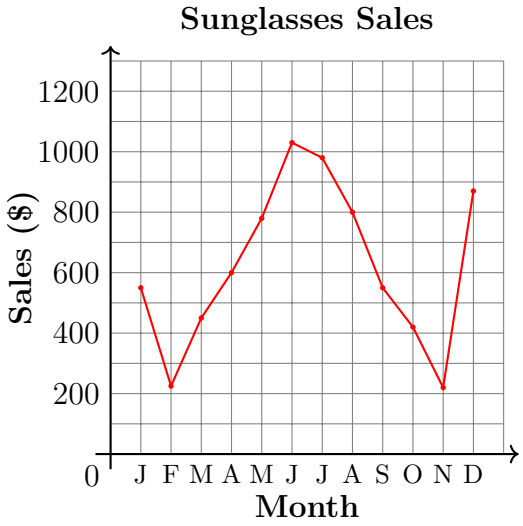
    - Which sampling methods often produce **invalid conclusions**?
    - Why are these problematic methods **still used**?

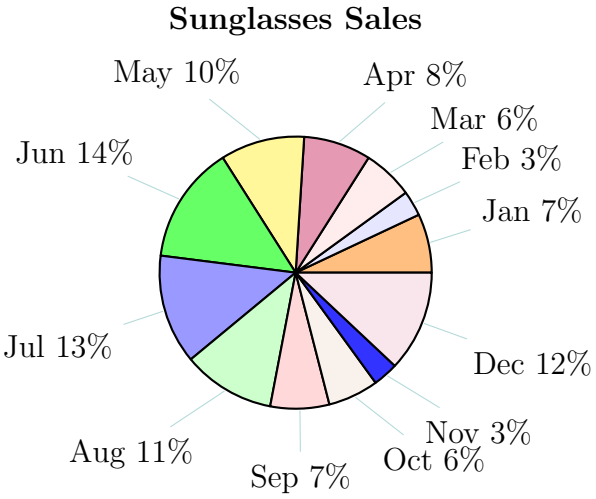
3 Representing Data

When there is a lot of data, we often choose a method to represent it that is easier to read or digest. There are often many ways to do this, and some can be misleading (unintentionally or intentionally).

The below three graphs represent the same data set.







Example 3.1 What are some unique features of each graph above?

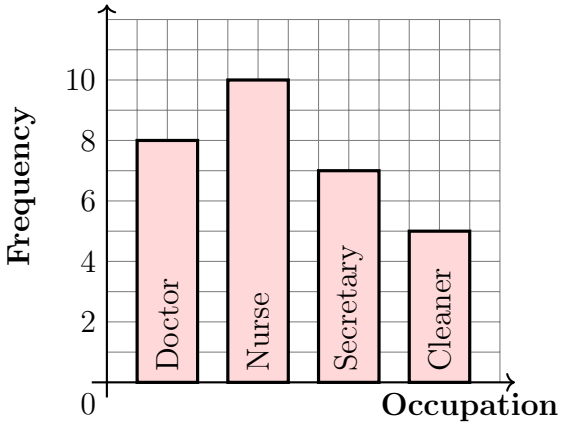
Different Types of Data

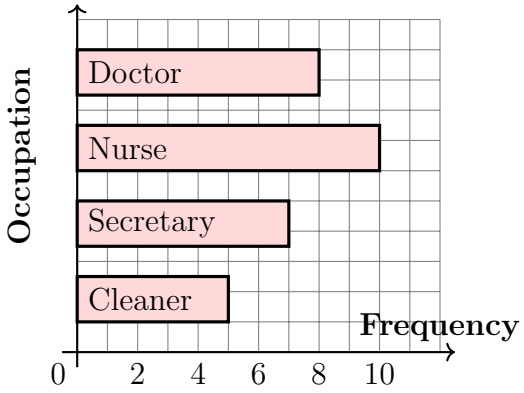
Categorical Data

Data that describes a particular quality or characteristic. Not numerical. Some examples include, genres, jobs, subjects, etc...

Likely a bar or column graph to be most suitable.

Example 3.2 Below are graphs that represent a survey of different jobs at a hospital.

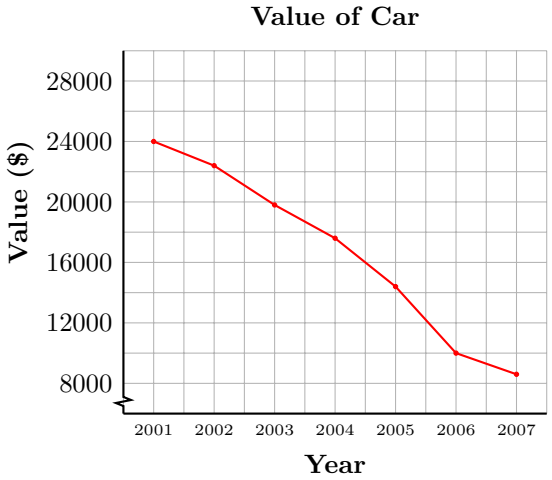


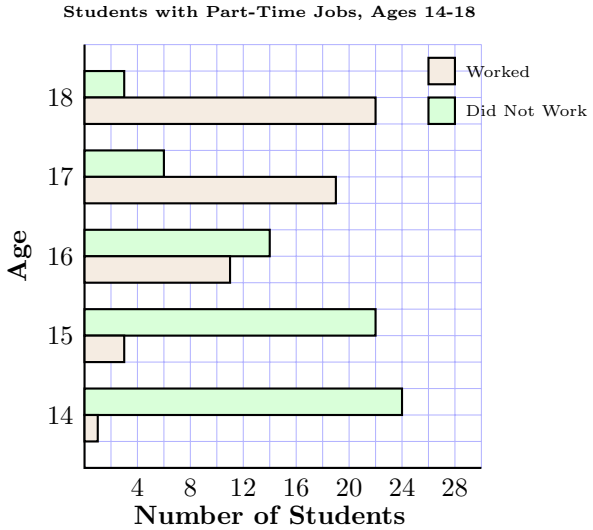


Numerical Data

Data that is given in numerical form and can be further classified as **discrete** or **continuous**

- **Discrete** - can only take distinct values and is usually a result of counting.
- **Continuous** - can take any value in a certain range and is usually a result of measuring.





Types of Graphs

Depending on the type of data, certain graphs may be better suited than others.

- For categorical data, a **bar/column graph** is a good choice
- For continuous numerical data, a **line graph** is a good choice
- For comparison to the whole data set, a **pie/circle graph** is a good choice
- **Pictographs** are not usually used, unless the data is simple and you want a potentially more engaging visual

Constructing a Circle Graph

Circle graphs are quite unique, because they display part of the whole. In some scenarios, this is a great visual benefit, but it also hides certain data at times. In addition, they are also terribly difficult to draw by hand. It is much more sensible to use technology for all graphs, and in particular, circle graphs.

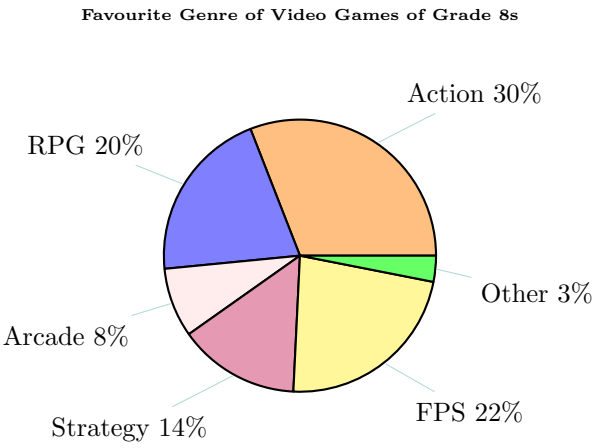
We need to make several calculations before we can understand how to construct a circle graph.

$$Whole\,Data\,Group = 100\% = 360^\circ$$

Each data that is part of the whole will then be converted to a percent and then calculated as a degree to find out the sector size.

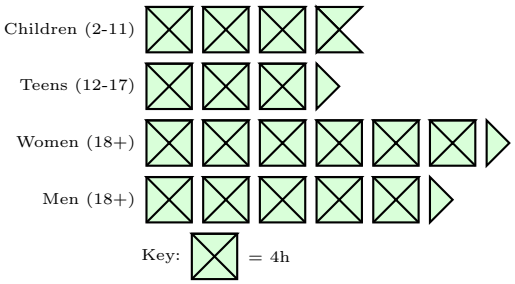
Example 3.3 Calculate the degree represented by each genre of video game.

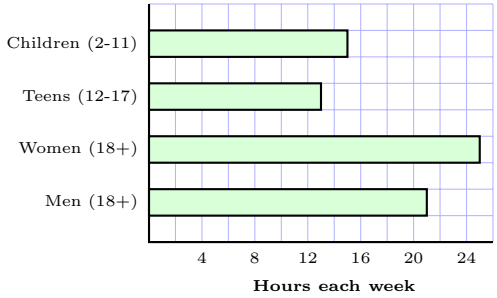
Type	Number of Students	Fraction	Percent	Degree
Action	15	$\frac{15}{50} = \frac{3}{10}$	30%	
RPG	10	$\frac{10}{50} = \frac{1}{5}$	20%	
Arcade	4	$\frac{4}{50} = \frac{2}{25}$	8%	
Strategy	7	$\frac{7}{50}$	14%	
FPS	11	$\frac{11}{50}$	22%	
Other	3	$\frac{3}{50}$	6%	



Practice Exercises

1. Each graph below shows how much time Canadians spend watching TV each week.





a) List 3 things you know from the bar graph.

b) List 3 things you know from the pictograph.

c) Which graph is more appropriate to display the data? Justify your choice.

2. You have the following surveyed data for the number of times a grade 8 student littered in a school week.

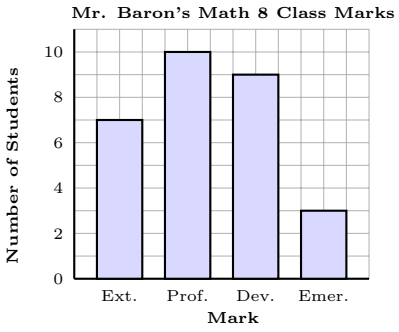
Litter Frequency	Number of Students
0	9
1-3	10
4-6	5
7+	2

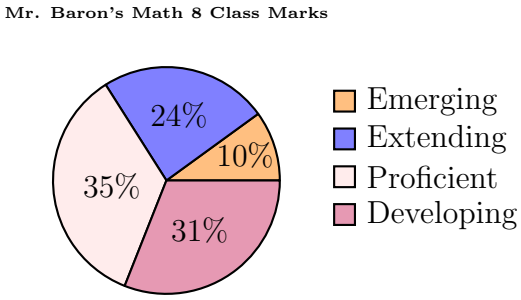
a) What type of graph is most appropriate for this data set?

b) Why might someone choose to use a pictograph for this data set?

c) Why might someone choose to use a circle graph for this data set?

3. The graphs below show the final marks for Mr. Baron's Grade 8 math class.



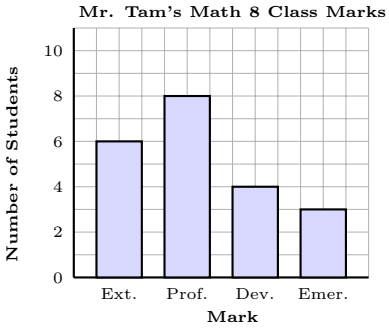


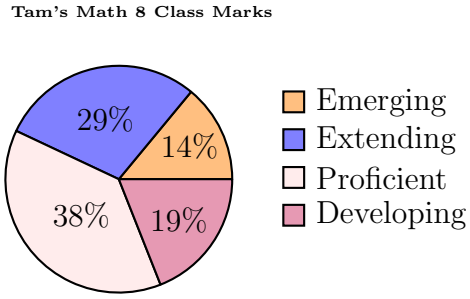
a) List 3 things you know from the column graph.

b) List 3 things you know from the circle graph?

c) Which graph best shows the number of students who got 'proficient' as a final grade? Justify your choice.

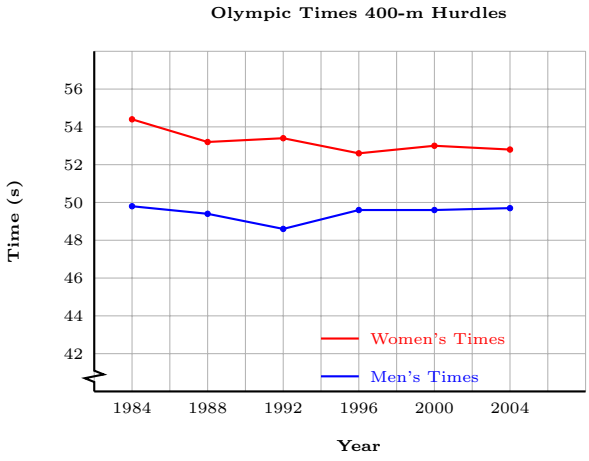
d) Below are the graphs for Mr. Tam's Grade 8 math class.

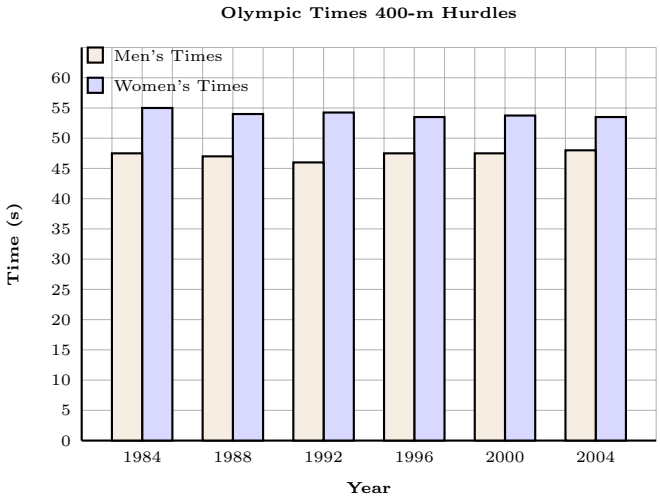




Which class do you think did better? Why do you think so?

4. Use the graphs below to answer the following.





a) What data do the graphs represent?

b) What is an advantage of each graph?

c) What is a disadvantage of each graph?

d) Which graph would you choose in each case? Explain your choice.

i) You want to show how the times changed over time.

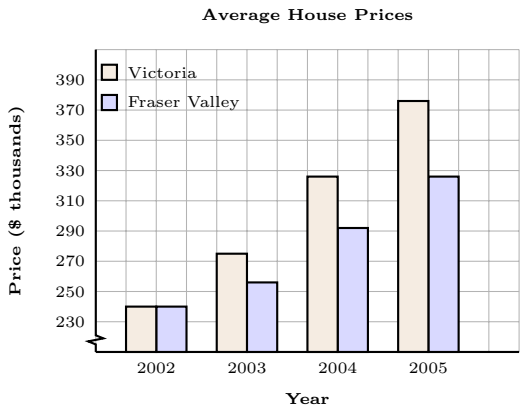
ii) You want to show the differences between the times for each year.

4 Misrepresenting Data

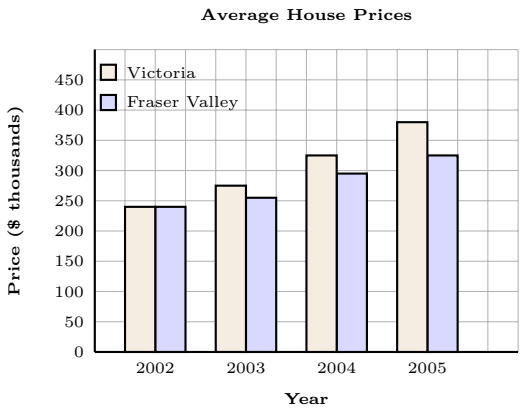
It is rather easy to misrepresent data and there needs to be a conscious effort to not make this mistake. Thankfully, technology will often just summarize the data completely and be rather unbiased. However, there are certain times where it makes sense to adjust the graph a bit, especially when dealing with a wider range of numbers.

Example 4.1 Compare the two graphs below. Is one misleading? Explain.

a)

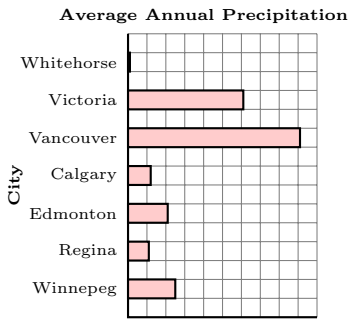


b)

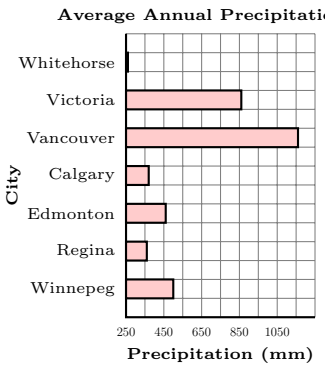


Example 4.2 Critique each of the graphs. Identify any misrepresentations.

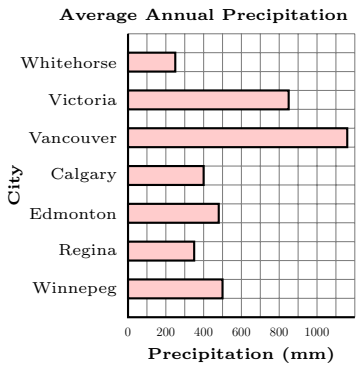
a)



b)



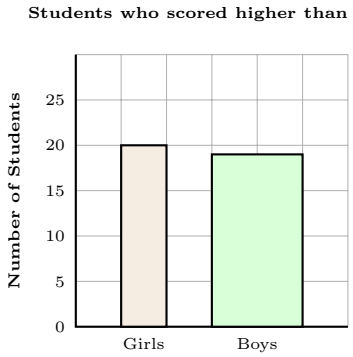
c)



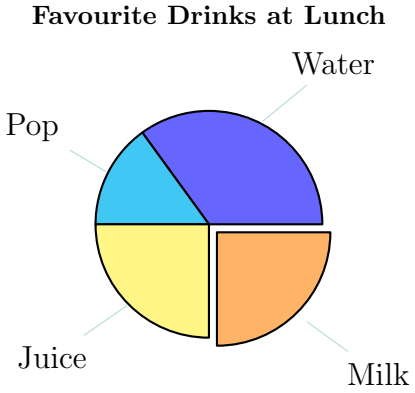
Data can also be misrepresented by purposely changing certain visual aspects of the graph.

Example 4.3 Critique each of the graphs below.

a)



b)

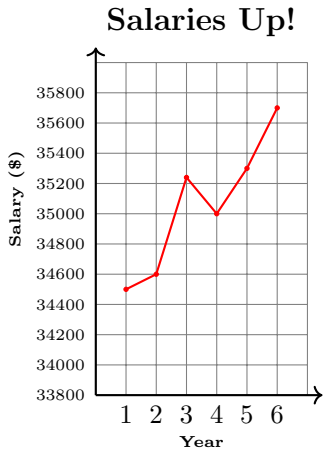


Scaling

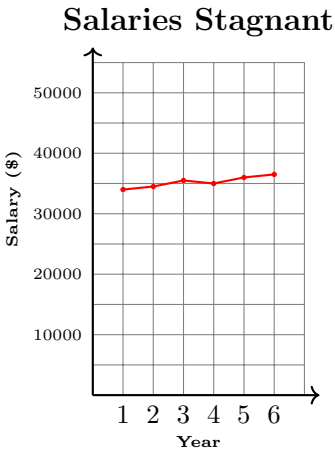
Choosing an appropriate scale when making graphs is important, otherwise, it can lead to misinformation or unintentional bias.

Example 4.4 Critique the scaling of each line graph below.

a)



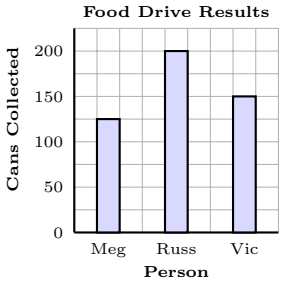
b)



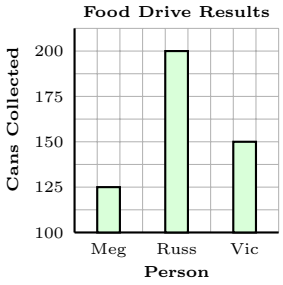
Practice Exercises

1. These graphs display the same data. Which graph is misleading? Why?

a)

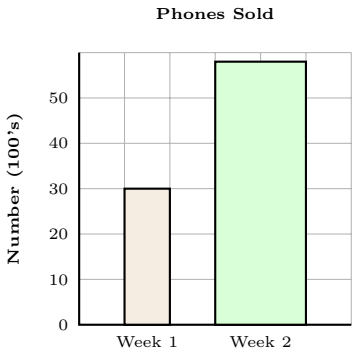


b)

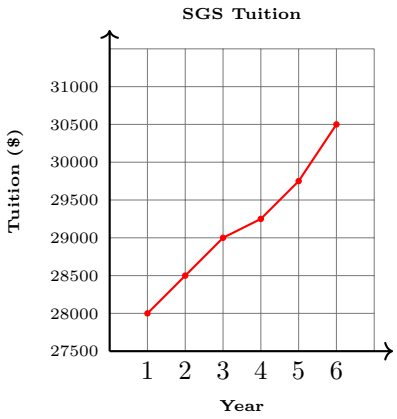


2. Describe the misleading or poor features of each of the following graphs. Then explain how you would fix each graph so it is not misleading

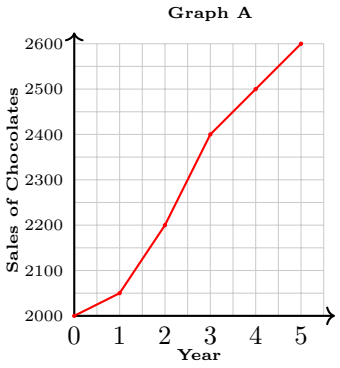
a)

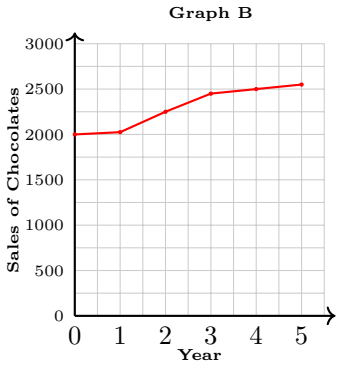


b)



3. Use the two graphs below to answer the following



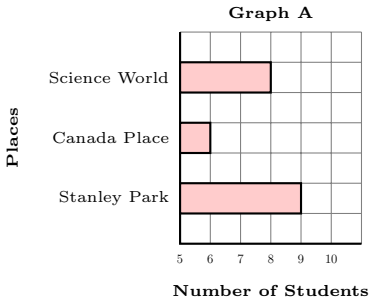


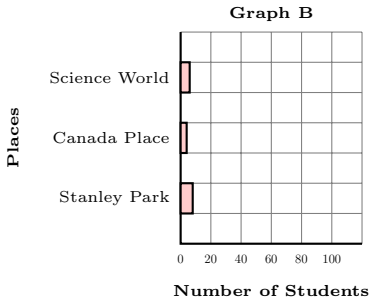
a) Which graph gives the impression of rapidly increasing sales? Explain.

b) Have sales in fact rapidly increased over this 5 year period? Explain.

c) According to graph A, the sales in year 3 appear to double the sales in year 2. Is this true? Explain.

4. A Grade 8 class in Vancouver was surveyed to find out where they would like to go for an end-of-year trip. The results where graphed by two students.





a) How is each graph misleading?

b) What changes would you make to graph these data accurately?

5 Probability

Calculating Probabilities

When all outcomes are equally likely, the probability of an event is:

$$\frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

- The result is always between 0 (impossible) and 1 (certain).
- Can be written as a fraction, decimal, or percentage.

It is important to know some normal expected values of common events.

Coins

A fair coin has 2 equally likely outcomes: Heads (H) or Tails (T).

Example 5.1 What is the probability of getting Heads?

Dice

A standard 6-sided die has outcomes: 1, 2, 3, 4, 5, 6. Dice is plural for die.

Example 5.2 What is the probability of rolling a 4 on a six-sided die?

Example 5.3 What is the probability of rolling an odd number on a six-sided die?

Playing Cards

A standard deck has 52 cards: 4 suits (Hearts, Diamonds, Clubs, Spades) with 13 ranks each (Ace–King).

Face cards are Jack, Queen, and King. Colours are red (Hearts and Diamonds) or black (Clubs and Spades).

Example 5.4 What is the probability of drawing an Ace in a standard deck of playing cards?

Example 5.5 What is the probability of drawing a face card that is spade?

Probability with Multiple Cases

When calculating probabilities, certain keywords indicate that multiple favourable outcomes must be considered. These include:

- **At least** (minimum value or more)
- **At most** (maximum value or less)
- **A or B** (either event occurring)
- **Greater than/Less than** (value comparisons)

Example 5.6 Find the probability of the following.

a) Rolling a number greater than 2 on a six-sided die

b) Selecting a 4 or a face card at random from a regular playing deck.

Probability Notation

The standard notation for probability uses $P(E)$ to represent the probability of event E occurring.

Example 5.7 Let E be the event that we draw a red ten from a regular deck of cards. Find $P(E)$.

Complement Events

For any event E, its complement E' represents "E does not occur".

Example 5.8 Let E be the event that we roll a prime number from a regular six-sided die. Find $P(E)$ and $P(E')$.

Complement Probability Rule

The fundamental relationship between an event and its complement is:

$$P(E) + P(E') = 1$$

This rule has several important implications:

- If $P(E)$ is known, $P(E') = 1 - P(E)$
- Particularly useful when $P(E')$ is easier to calculate than $P(E)$

Example 5.9 Find the probability of drawing a card from a regular playing deck that is not a jack or a red 6.

Sample Space

The **sample space** (denoted S) of an experiment is the set of all possible distinct outcomes that can occur.

Experiment	**Sample Space** S
Single coin flip	$\{H,\ T\}$
Rolling a 6-sided die	$\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$
Drawing a card from a deck	{Ace♡, 2♡, ..., King♠} (52 elements)
Two coin flips	$\{HH,\ HT,\ TH,\ TT\}$

Example 5.10 Given the sample space below of rolling two dice. Answer the following

	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Example 5.11 Let E be the event that we roll two dice with identical numbers. Find $P(E)$.

Example 5.12 Let E be the event that we roll two dice with a total sum of 8. Find $P(E)$.

Example 5.13 Let A be the event that we roll two dice with a total sum less than 6. Find $P(A)$.

Example 5.14 Let B be the event that we roll two dice with two numbers that differ by exactly 2. Find $P(B)$.

Compound/Consecutive Events

This is a key concept in probability. In general, the probability of consecutive **independent** events happening, is the product of their individual probabilities.

$$P(A\,and\,B) = P(A) \times P(B)$$

Example 5.15 Find the probability of correctly guessing 5 multiple choice questions in a row (A, B, C, D as options).

Example 5.16 Find the probability of drawing 3 consecutive Aces in a well-shuffled deck of cards (with replacement).

Example 5.17 Steve Nash is a career 90% free throw shooter. What is the probability that he will get in 10 consecutive free throws?

Independent and Dependent Events

We define **independent events** as events that do not affect the occurrence of the other. For example, flipping a coin and drawing a playing card are considered independent.

We define **dependent events** as events that do affect the occurrence of the other. For example, drawing two consecutive Aces *without replacement*.

If A and B are independent, then $P(A\,and\,B) = P(A) \times P(B)$

If A and B are dependent, then $P(A\,and\,B) = P(A) \times P(B\,given\,A\,has\,occured)$

Example 5.18 You have a bag of candy that consists of 10 KitKats and 14 Aeros. What is the probability of randomly drawing two KitKats, without replacement? With replacement?

Tree Diagrams

A **tree diagram** is a visual tool to represent:

- Sequential events (e.g., multi-step experiments)
- All possible outcomes and their probabilities

Example 5.19 Construct a tree diagram to show the outcomes and probabilities of flipping a coin three times.

Practice Exercises

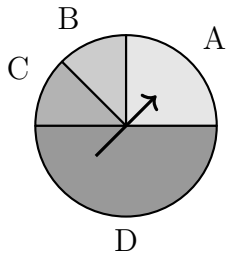
1. If an event is certain, what is its probability?

2. If an event cannot happen, what is its probability?

3. What is the sum of all the probabilities in a sample space?

4. If the probability that there will be a quiz is $\frac{1}{4}$, what is the probability that there will not be a quiz?

5. If the arrow is spun, determine the probability of it stopping in the specified regions.



a) $P(A)$:

b) $P(B)$:

c) $P(C)$:

6. A fair die is rolled, find the following.

   a) $P(3)$

   b) $P(\text{an even number})$

   c) $P(\text{an odd number less than 4})$

   d) $P(\text{not a 5})$

7. Two fair dice are rolled, find the following.

   a) $P(\text{sum of 7 or 11})$

   b) $P(\text{sum of 6})$

   c) $P(\text{doubles})$

   d) $P(\text{sum greather than or equal to 9})$

8. A marble is drawn from a bag containing 8 black, 5 white, 4 green, and 3 red marbles. Find the following.

   a) $P(\text{black})$

   b) $P(\text{white or red})$

   c) $P(\text{not green})$

   d) $P(\text{white or green or red})$

9. A deck of 52 cards from Ace to King has four suits: Hearts and Diamonds, which are red, and Spades and Clubs which are black. Jacks, Queens, and Kings are called face cards. Determine the following probabilities:

   a) Selecting a face card

   b) Selecting a heart

   c) Selecting a red card

   d) Selecting a black face card

   e) Selecting a club or diamond

   f) Not drawing a spade

   g) Not drawing a black face card

   h) Not drawing a red 10

   i) Drawing a 7, 8, or 9

   j) Drawing a face card, or an Ace, or a card less than 6

   k) Drawing an even number red card

   l) Drawing an even numbered diamond

   m) Drawing a 5 or a club

   n) Drawing a face card or a heart

10. State if the two events are dependent or independent

    a) Tossing a coin and picking a card

    b) Tossing a coin twice

    c) Picking two balls from a bag

    d) Picking a ball from a bag, replacing the ball, then picking again.

11. A coin is flipped three times. What is the probability that the coin lands tails all three times?

12. If 80% of the students in a math class get an A, what is the probability that two students at random both got A's in math?

13. If two people are selected at random, what is the probability that they were both born in October?

14. If two people are selected at random, what is the probability that they were both born in the same month?

15. A coin is flipped 4 times. Determine the probability of the coin landing heads at least once.

16. A die is tossed 4 times. Determine the probability of getting at least one 5.

6 Reasoning and Analyzing Problems

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 6.1 For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) The mean of a data set is larger than the median of the data set.

b) The median of a data set is a value in the data set.

c) Each data set can have at most one mode.

d) If $P(A) = 0.6$, then $P(A') = 0.4$.

e) If $P(A) = 0$, then A is an event that is impossible.

f) If $P(A) = m$ and $P(B) = n$, then $P(A \, and \, B) = mn$.

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) The median of the data set is larger than the mode of the data set.

   (b) The mean of the data set is smaller than the largest value.

   (c) A line graph is a suitable representation for discrete numerical data.

   (d) Let A be any event, then $0 \leq P(A) \leq 1$.

   (e) If $P(A \, and \, B) = P(A) \times P(B)$, then A and B are independent events.

   (f) $P(A)$ can be negative, as long as $P(A) + P(A') = 1$.

Answer Key

Section 1 - Exercises

1. a) $n = 23$, mode $= 6$, median $= 6$, mean $= \frac{129}{23} \approx 5.6$
   b) $n = 15$, mode $= 18$, median $= 17$, mean $= \frac{245}{1} \approx 16.3$
   c) $n = 11$, mode $= 23.5$, median $= 24.9$, mean $= \frac{273}{11} \approx 24.8$
   d) $n = 21$, mode $= 115, 127$, median $= 128$, mean $= \frac{2700}{21} \approx 128.6$

2. a) $n = 10$, mode $= \$33,000$, median $= \$43,500$, mean $= \$42,800$
   b) The mode is the lower value in the data set.

3. a) 19 mm b) 19 mm on Saturday c) Approx. 17.7 mm

4. a) median: \$10000 mode: \$10000 mean: \$100000
   b) Answers may vary. If negotiating with a foreman, I would use the mode/median to say that "most" salaries are low and that the foreman's salary is already very high.
   The mean could be used to argue that the average salary is very high and that there cannot be wage increases.
   c) Answers may vary. Use mode/median to compare against the mean to show that the majority of salaries are lower than the average, so most people are earning less than the average.

5. a) mode b) mean c) median d) mean

6. 13.6

7. 21

8. 37.5

Section 2 - Exercises

1. ANSWERS MAY VARY.
   a) The survey is biased by only providing Cola as options.
   b) Already suggesting the Prime Minister is the best, forcing respondent to be defensive.
   c) Privacy and ethics. People may not want to disclose this information as they don't know what it will be used for.

2. ANSWERS MAY VARY.
   a) Sample size is too small, friends might share similar opinions. Not a good representation of the whole school.
   b) Ethical decision. People may be inclined to choose agree, even if not true because of expectations of being moral and not to be judged.
   c) Cultural sensitivity. Not being aware of different cultures and openly criticizing it. People may not feel responding.
   d) This would be like a census, so it will take too long.

3. ANSWERS MAY VARY.
   a) wording of 'violent criminals' shows bias
   b) wording of 'gas guzzling' shows bias
   c) wording of 'automatically improves' shows bias

4. ANSWERS MAY VARY.
   a) Should DNA tests be used in court as evidence?
   b) Should SUVs be banned?
   c) Should students use spell check?

5. Answers may vary. Some topics could be political support, age, drug use, sex, etc...

6. a) Census b) Survey

7. a) People that use the bus b) All customers that pay GST
   c) Parents with children that need daycare. d) Patients that have been to the emergency room before.

8. Answers may vary.
   a) Convenience sampling, not a good representation. Highly biased.
   b) Students should also be surveyed.
   c) Other age groups should be surveyed.

9. Answers may vary
   a) depends on the audience and if they represent the town adequately, but this could be appropriate.
   b) appropriate for the day, but restaurants vary greatly dependent on days, so polling on different days is important.
   c) this seems fair. 50 is a decent sample size of the school and if the policy is for only students, then this is appropriate.
   d) if your sample does not include cyclists, then the sample is not representative of what you're trying to survey

10. Answers may vary.

    a) Stratified random: certain number from each province.

    b) Systematic, stratified random, or random.

11. - Cluster, convenience sampling can often produce invalid conclusions due to high chance of bias.
    - convenience and time/cost effectiveness are probably the main reasons

Section 3 - Exercises

1. Answers may vary.

    a) Women watch the most tv, teens watch the least, children watch more than teens,...

    b) children watch 2 more hours than teens, women watch four hours more than men, teens are the least at 13 hours

    c) If counting by whole hours only, the pictograph is nice. Buf if fractional hours are allowed, then the bar graph is better.

2. Answers may vary.

    a) bar graph or column graph

    b) data is whole numbers, no fractions, could be more engaging

    c) to see comparison to the whole data set, or what percentage of students litter at different frequencies.

3. Answers may vary.

    a) very few emergings, mostly proficient and developing, 29 students in total

    b) the percentages of students for each level, developing level is triple emerging, extending is about 2.4x emerging

    c) Column graph is best because we can see exact numbers.

    d) Percentage wise, Mr. Tam's class can argue they did better as there are higher percentages of Extending/Proficient. Column graph shows that there are fewer students in Mr. Tam's class though, so Mr. Baron would have more extending/proficient students by number.

4. Answers may vary.

   a) men and women's hurdle times in the Olympics from 1984-2004

   b) line graph shows a more noticeable change in times, uses the line break
   bar is more true to show the change is not as drastic or that the difference between men and women are not as big.

   c) line graph may show a bias to represent men are significantly faster
   column graph has a long going on, might be less appealing. Scale maybe is too large at 5.

   d)

      i) line graph, shows trend by lines

      ii) column graph shows a nice difference in time between the genders and can be read relatively easily

Section 4 - Exercises

1. Answers may vary. Graph b) is misleading because the scale is not correctly done, appears that Vic brought in twice as many cans as Meg.

2. Answers may vary.

   - Area of week 2 is way bigger, not the same scale.
   Use the same width for each week.
   - No jump break, scale goes up by $500 which shows a much steeper increase over time.
   Use a jump break and a more appropriate scale.

3. Answers may vary.

   a) Graph A shows a much faster growth.

   b) Not really, unless $600 is considered a huge increase.

   c) No, the vertical scale is off and no line break was used to show it was adjusted.

4. Answers may vary.

   a) A - not starting at 0
   B - scale is too large, showing very little difference on graph.

   b) Start from 0 as number of students, and go up by 2's

Section 5 - Exercises

1. 100% or 1
2. 0% or 0
3. 100% or 1
4. $\frac{3}{4} = 0.75 = 75\%$
5. a) $\frac{1}{4} = 0.25 = 25\%$ b) $\frac{1}{8} = 0.125 = 12.5\%$ c) $\frac{1}{8} = 0.125 = 12.5\%$
6. a) $\frac{1}{6}$ b) $\frac{1}{2}$ c) $\frac{1}{3}$ d) $\frac{5}{6}$
7. a) $\frac{2}{9}$ b) $\frac{5}{36}$ c) $\frac{1}{6}$ d) $\frac{5}{18}$
8. a) $\frac{2}{5}$ b) $\frac{2}{5}$ c) $\frac{4}{5}$ d) $\frac{3}{5}$
9. a) $\frac{3}{13}$ b) $\frac{1}{4}$ c) $\frac{1}{2}$ d) $\frac{3}{26}$
   e) $\frac{1}{2}$ f) $\frac{3}{4}$ g) $\frac{23}{26}$ h) $\frac{25}{26}$
   i) $\frac{3}{13}$ j) $\frac{8}{13}$ k) $\frac{5}{26}$ l) $\frac{5}{52}$
   m) $\frac{4}{13}$ n) $\frac{11}{26}$
10. a) independent b) independent c) dependent d) independent
11. $\frac{1}{8} = 0.125 = 12.5\%$
12. $0.64 = 64\%$
13. $\frac{1}{144}$
14. $\frac{1}{12}$
15. $\frac{15}{16}$
16. $\frac{671}{1296} \approx 51.8\%$

Section 6 - Exercises

1. a) Sometimes true. 1,1,3,4,5 has median 3 and mode 1. However, 2,4,5,6,6 has median 5 and mode 6.

   b) Sometimes true. 9,9,9 has mean 9 and largest value 9. 1,2,3 has mean 2 and largest value 3.

   c) Never true. Discrete numerical data means they are not continuous, so a line graph would be a poor choice.

   d) Always true. Probability must always be between 0 and 1 inclusive.

   e) Always true. By definition, if $P(A\&B) = P(A)\times P(B)$, then they are independent.

   f) Never true. $P(A)$ cannot be negative as $0 \leq P(A) \leq 1$.